

# KNEE ARTHROCENTESIS & INJECTION — One-Pager

Superolateral intra-articular approach + knee exam · TeachIM simulation curriculum · IM resident reference

## 1 · INDICATIONS

- Symptomatic knee OA; aseptic inflammatory arthritis flare
- Diagnostic aspiration: evaluate effusion, r/o septic joint / crystals

## 2 · CONTRAINDICATIONS

|                         |                                                                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Absolute</b>         | <b>Hardware/prosthesis in joint;</b> overlying skin infection; <b>suspected septic joint</b> (aspirate & treat, do NOT inject steroid) |
| <b>NOT contraindic.</b> | Anticoagulation; well-controlled DM; surgery >3 mo out (defer steroid if <3 mo)                                                        |

## 3 · KNEE EXAM (+LR)

| Finding            | Suggests         | +LR  |
|--------------------|------------------|------|
| Bony enlargement   | OA               | ~12  |
| Varus deformity    | OA               | ~3.4 |
| Stiffness <30 min  | OA               | ~3   |
| Crepitus           | OA               | ~2   |
| McMurray's +       | Meniscus         | ~4   |
| Can't fully extend | Meniscus (mech.) | ~3.2 |

**Pes anserine bursitis:** point TTP anteromedial tibia ~2–5 cm BELOW the joint line (anterior & inferior to it) — distinguishes from medial meniscus tear.

## 4 · WHAT WORKS (ACR 2019)

- **Core:** exercise/PT + weight loss — **more effective long-term** than injection
- **First-line drug:** topical NSAID (diclofenac), oral NSAID
- **Steroid injection:** small-mod relief ~6 wk (Cochrane 2015) — for flares/short-term
- ACR recommends **AGAINST** hyaluronic acid, PRP, stem cells

**Don't over-inject:** repeated IA steroids may accelerate cartilage/structural loss (McAlindon JAMA 2017; OAI 2019).

## 5 · TECHNIQUE (superolateral)

- **Position:** supine, leg **fully extended** (relaxes quad/patellar tendon).
- Mark superior + lateral patellar edges → insertion point **superolateral to patella**.
- Chlorhexidine scrub ×30 s. Displace patella laterally.
- Needle **parallel to floor**, slightly under patella (perpendicular to leg or slightly toward toes), ~3 cm.
- **Aspirate first** (no blood) → inject lido + triamcinolone 40 mg **WITHOUT resistance**.

**Resistance = NOT in joint space** → withdraw & redirect.  
Free flow = intra-articular.

## 6 · COUNSELING

- **Diabetics:** glucose may rise >100 mg/dL for ~24–48 h (up to several days), worse with poor baseline control.
- Refer PT; start within ~4 wk for durable benefit.
- Return precautions: signs of joint infection.

## MENISCAL TEAR: PT vs SURGERY

- **Degenerative tear (older + OA):** PT = arthroscopy for pain/function (Khan CMAJ 2014); MRI rarely changes mgmt.
- **Surgery favored:** younger, acute tear, true mechanical sx (locking / can't extend).

## SUPPLIES

Marker · chlorhexidine · 1% lido (5 cc) · triamcinolone 40 mg/mL (1 cc) · 18 g (draw-up) + 21–25 g 1.5" (inject) · 5 cc syringe · gauze · band-aid (+ 18 g & 20–30 cc syringe if aspirating).

Adapted from TeachIM.org Knee Arthrocentesis (Fainstad, Brett, Thompson, Melton; Feb 2021, rev. 2023/2025), CC BY-NC 4.0 — non-commercial educational use. Optimized June 2026; currency verified (ACR/AF 2019 OA guideline current; McAlindon JAMA 2017 progression caution). Clinician-review before teaching. Not medical advice.